CERTIFICATE IN INFORMATION AND COMMUNICATION TECHNOLOGY FOR THE VISUALLY IMPAIRED

CHART

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CURRICULUM DEVELOPMENT AND ADVISORY SERVICES
TRAINING STANDARDS DIVISION
TEVETA

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1. RATIONALE

The most commonly accessed career by the visually impaired (VI) in Zambia has been telephone operating, not disregarding teaching, music and a few other careers. However, the employment opportunities of the VI have continued to dwindle due to advances in technology, for example the introduction of digital telephone exchange, musical instruments and teaching aids. It is also commonly known that the VI have had difficulties in their studies because most literature is in print form.

The VI in Zambia had their first exposure to training in handicrafts in the late 1920s. After that, several other technical training programmes were developed with the support of missionaries. After independence in the late 1960s, Saunders was contracted by the Zambian Government to carry out a situation analysis of Technical Education in Zambia. Following his report in 1968, Technical Education was officially introduced. Thereafter, several training programmes were developed to meet the technical labour demands of industry. In 1999, the first nine visually impaired persons were introduced to basic computer skills in a two-week workshop sponsored by KEPA Zambia.

Through the years, various developments have been made in the field of technology, broadening opportunities for the VI in the areas of education, employment and entrepreneurship. With the latest technological developments in such areas as Information and Communication Technology (ICT), the VI can increasingly study on their own, widen their knowledge and skill base, and expand their employment and entrepreneurship opportunities.

In consideration of the above, TEVETA has taken the initiative to develop a training programme leading to Certificate in Information and Communication Technology for the Visually Impaired. This training programme has been presented in a modular form integrating both theoretical and practical training. It is hoped that by implementation of this programme, TEVETA will contribute to the creation of an inclusive industrial society, which will be less discriminatory to the visually impaired.

2. ACKNOWLEDGMENT

TEVETA wishes to express sincere appreciation to each member of the Curriculum Development and Review Team for the helpful and constructive criticism of the course content of the Information and Communication Technology for the visually impaired programme.

The curriculum has been produced with the assistance of the Curriculum Development Review Team comprising the following people:

1. Mr. Adrian S Chikapa — Zambia Institute for Special Education (ZAMISE)
2. Mr. Frank Chikwale — Examinations Council of Zambia (ECZ)
3. Mr. Michael M Chilala — Touch for Progress
4. Mr. Kosamu Daka — KDZ Enterprises
5. Dr. Mwamba D Kalabula — University of Zambia (UNZA)
6. Mr. Musonda Kombe — Touch for Progress
7. Mr. Edgar Mbewe — Touch for Progress
8. Mr. Hamanyanga Miyoba — Touch for Progress
9. Mrs. Agnes M K Tembo — National Vocational Rehabilitation Centre (NVRC)
10. Mr. Orphan Hachinene — TEVETA
11. Mr. Gabriel S Konayuma — TEVETA
12. Miss. Hazel Njoloma — TEVETA

3. PROGRAMME GOAL

This programme is designed to equip the visually impaired with skills in Information and Communication Technology, Braille production and entrepreneurship to meet the needs of industry.

4. PROGRAMME AIMS

This programme aims at providing trainees with knowledge and skills in:

5.1 Installation of computers
5.2 Computer operations
5.3 Information Technology in relation to Braille production
5.4 Internet and e-mail
5.5 Information Technology in relation to telephone exchange operation
5.6 Communication
5.7 Entrepreneurship

5. PROGRAMME OBJECTIVES

On completion of the course, the visually impaired student will be able to:

5.1 Install and use an operating system
5.2 Install and use Computer application packages

5.3 Install, use and apply skills in Internet and intranet
5.4 Install, use and apply skills in computerised Braille
5.5 Operate a computer based telephone exchange
5.6 Apply skills in entrepreneurship

6. ENTRY REQUIREMENTS

GCE "O" Level with a pass in English Language.

7. ASSESSMENT

7.1	**CONTINUOUS ASSESSMENT:**	**40%**
	7.1.1 Practical:	20%
	71.2 Theory:	20%
7.2	**FINAL EXAMINATIONS:**	**60%**
	7.2.1 Practical:	40%
	7.2.2 Theory:	20%

8. MINIMUM PASS LEVEL

The minimum pass level shall be **50%** for each module.

9. REFERRALS AND FAILURES

9.1 Students who fail four (4) modules shall be considered to have completely failed.
9.2 Students who fail three (3) modules shall be referred and given a maximum of two chances to re-sit the modules.
9.3 Students who fail one (1) module shall proceed but re-sit the module they have failed.

10. ATTENDANCE

80% of teaching time.

11. STAFF QUALIFICATIONS

Trainers shall have a minimum qualification of a Diploma in a relevant field with teaching qualifications from a recognised institution.

12. COURSE DURATION

Two semesters of four months each and two months industrial attachments. The total course duration will be 1042 hours.

13. DELIVERY METHODS

A combination of methods: lecture, practical, and tutorials.

14. PROGRAMME EVALUATION

The purpose of programme evaluation is to determine on an on-going basis whether it is being implemented as planned and to get feedback from industry on whether the programme meets industrial demands.

14.1 FORMATIVE EVALUATION

14.1.1 MAJOR AREAS OF EVALUATION

Course aims and objectives, student's entry requirements, course content, and teaching/learning resources.

14.1.2 EVALUATION INSTRUMENTS

Questionnaires, interviews, observations, checklists, examination/ test records.

14.1.3 SOURCES OF INFORMATION

Students, lecturers, administrators, drop outs and the Examining Board.

14.2 SUMMATIVE EVALUATION

14.2.1 MAJOR AREAS OF EVALUATION

Course aims and objectives, student's entry requirements, course content, teaching/learning resources, teaching /learning activities, graduate performance in industry.

14.2.2 EVALUATION INSTRUMENTS

Questionnaires, interviews, observations, checklist, examination/ test records.

14.2.3 SOURCES OF INFORMATION

Students, lecturers, administrators, dropouts, Examining Board, employers, unions, committees, and general public.

15. CERTIFICATION

TEVETA shall award a certificate in ICT for the Visually Impaired to successful candidates. The grading is as follows:

50% - 59%	Pass
60% - 69 %	Credit
70% - 84%	Merit
85% - 100%	Distinction

MODULE 1 COMMUNICATION SKILLS

Duration: **140 HOURS**

Aim: To equip trainees with skills and knowledge for effective and efficient communication**.**

Objectives: At the end of the module the trainee will be able to communicate effectively and efficiently using:

1. The Principles of Communication
2. Oral Communication
3. Written Communication
4. Visual Communication
5. Electronic Communication

Unit 1.1 Principles and Process of Communication

1.1.1 Nature and purpose of communication
1.1.2 Communication skills
1.1.3 Communication process strategies
1.1.4 Steps to effective communication
1.1.5 Barriers to effective communication

Unit 1.2: Oral Communication

1.2.1 Effective listening skills
1.2.2 Effective questioning skills
1.2.3 Note taking skills
1.2.4 Effective public presentations and speaking skills
1.2.5 Organizing meetings
1.2.6 Interviewing as a communication tool

Unit 1.3 Written Communication

1.3.1 Types of written communication
1.3.2 Drafting methods
1.3.3 Types of reports
1.3.4 Characteristics of good report writing
1.3.5 Report writing activity related to IT field

Unit 1.4 Visual Communication

1.4.1 Principles of Visual Communication
1.4.2 Basic components of Visual Communication
1.4.3 Design visual aids to advertise corporate identity and design

Unit 1.5 Electronic Communication

1.5.1 Modes of electronic communication
1.5.2 Internet communication methods

1.5.3 The use of emerging technologies in aiding business communication

Unit 1.6 Communication in Organisations

1.6.1 The role of communication in an organisation
1.6.2 Different organisation structures
1.6.3 Communication flows in different organisation structures
1.6.4 Decision making process in an organisation.
1.6.5 Describe methods of communication used in organisations.
1.6.6 Explain clearly the need for discipline and security procedures.
1.6.7 Explain the necessity of healthy and safety organisations.

RECOMMENDED READING

Bovee, C. L., Thill, J. V., Schatzman, B. E. (2002). *Business Communication Today 7th Ed.* New Delhi: Pearson Education Inc.

Swift & Stanwell (1986). *English in the Office,* London: Longman

Little, P (1987) *Communication in Business,* London: Longman

Gartside, L (1992) *Modern Business Correspondence,* London: Pitman

Kaul, A (2002) *Effective Business Communication,* New Delhi: Prentice-Hall

Taylor, S (1991) *Communication For Business,* London: Pitman

Stanton, S (1990) *Communication,* London: Longman

Wayne, F. S and Dauwalder, P. D (1994) *Communicating in Business,* New York: Irwin Inc

MODULE 2 **SCREEN REVIEW SOFTWARE AND INTRODUCTION TO COMPUTERS**

Nominal Duration: **210 hours**

Aim: The Module aims at providing Information Technology skills and competencies required in a real work situation.

Objectives: At the end of the Module, trainees will be able to:

1. Apply the Information Technology skills in Screen Review Software and Telephone Interface.
2. Describe Hardware and Telephone Interface.
3. Describe Hardware and Computer Classifications.
4. Describe Software applications
5. Describe the Information Processing Cycle
6. Use Storage Devices
7. Describe the Operational Characteristics of Storage Devices
8. Describe Operating Systems, Files and Directories

Unit 2.1 **Screen Review Software and Telephone Interface**

2.1.1 Definition of Screen Review Software and Telephone Interface
2.1.2 Types of Screen Review Software and Telephone Interface
2.1.3 Functions of Screen Review Software and Telephone Interface

Unit 2.2 **Hardware and Computer Classifications**

2.2.1 Three main classifications of computers.
2.2.2 Four functions of the information processing cycle.
2.2.3 Hardware components and their functions.
2.2.4 Types of computer storage devices.

Unit 2.3 **Software**

2.3.1 Key functions of systems software.
2.3.2 Widely used software applications.
2.3.3 Range of languages used in software development.
2.3.4 Organisation of data in software development.
2.3.5 New and emerging classes of software.

Unit 2.4 **Information Processing Cycle**

2.4.1 Input media.
2.4.2 Process phase.
2.4.3 Describe the various output media.
2.4.4 Describe the various storage media.

Unit 2.5 Storage Devices

2.5.1 Features and functions of a Floppy Disk Drive.
2.5.2 Features and functions of a Hard Disk Drive.
2.5.3 Features and functions of a RW CD-ROM Drive.
2.5.4 Features and functions of a Tape Drive.
2.5.5 Features and functions of a ZIP Drive.
2.5.6 Features and functions of a USB Flash Stick.

Unit 2.6 Operational Characteristics of Storage Devices

2.6.1 Operational functions and capacities of a Floppy Disk Drive.
2.6.2 Operational functions and capacities of a Hard Drive.
2.6.3 Operational functions and capacities of a RW CD-ROM Drive.
2.6.4 Operational functions and capacities of a Tape Drive.
2.6.5 Operational functions and capacities of a ZIP Drive.
2.6.6 Operational functions and capacities of a USB Flash Stick.

Unit 2.7 Operating Systems, Files and Directories

2.7.1 Definition of an Operating System.
2.7.2 Purpose of an Operating System.
2.7.3 Functions of an Operating System.
2.7.4 Factors to consider when choosing an Operating System.
2.7.5 Features of an Operating System.
2.7.6 Use and functions of files.
2.7.7 Use and functions of directories.

MODULE 3 APPLICATION PACKAGES

Nominal Duration: 182 hours

Aim: The module aims at developing Knowledge and Skills in the use of application packages.

Objective: At the end of the module the trainee will be able to:

1. Use Windows 98/ Windows 2000 Operating Systems
2. Use a Word Processing package e.g. Microsoft Word
3. Use a Spreadsheet package e.g. Microsoft Excel
4. Use a Word Processing package e.g. Microsoft Word
5. Use a Presentation Graphics Software package e.g. Microsoft PowerPoint
6. Use a Database package e.g. Microsoft Access
7. Use a Desktop Publishing package e.g. Microsoft Publisher
8. Send and Receive E-mail using Outlook Express and other e-mail accounts
9. Use the Internet using Microsoft Explorer and Netscape Navigator

Unit 3.1: Software: Windows 98/Windows 2000

3.1.1 Explain a Windows desktop

3.1.2 Start a computer

3.1.3 Shut down a computer

3.1.4 Open Windows Explorer window

3.1.5 Create a new folder in Windows Explorer

3.1.6 Delete a file or folder

3.1.7 Cut a file or folder

3.1.8 Paste a file or folder

3.1.9 Rename a file or folder

3.1.10 Select a screen saver

3.1.11 Explain the parts of a window

3.1.12 Switch between windows

3.1.13 Explain a dialog box

Unit 3.2: Word Processing: Microsoft Word

3.2.1 Identify the various parts of the MS-Word screen.

3.2.2 Create a simple document.

3.2.3 Use simple editing tools.

3.2.4 Save the document and exit.

3.2.5 Retrieve the document.

3.2.6 Format the document.

3.2.7 Use File menu.

3.2.8 Use the Edit menu

3.2.9 Use the View menu

3.2.10 Use the Tools menu.

3.2.11 Create Tables

3.2.12 Work with multiple documents

3.2.13 Use the Window menu

3.2.14 Get Help from MS-Word.

Unit 3.3: Spreadsheets: Microsoft Excel

3.3.1 Identifying the various parts of the MS-Excel screen.

3.3.2 Creating a simple worksheet

3.3.3 Using simple editing tools.

3.3.4 Saving the worksheet and exit.

3.3.5 Retrieving the worksheet.

3.3.6 Formatting the worksheet.

3.3.7 Using Tools menu

3.3.8 Using the Data menu

3.3.9 Using the View menu

3.3.10 Working with multiple worksheets

3.3.11 Using the Window menu

3.3.12 Getting Help from MS-Excel.

Unit 3.4: Presentation Graphics Software: Microsoft PowerPoint

3.4.1 Identify hardware items that support presentation software

3.4.2 Create computer presentation and handouts in accordance with basic principles of graphics design and visual communication

3.4.3 Edit presentations.

3.4.4 Copy from one presentation to other

3.4.5 Print single slide, an entire presentation, an outline and notes

3.4.6 Create clipart in a slide

3.4.7 Create / insert word art objects

3.4.8 Create or modify a graph on a slide

3.4.9 Use templates in presentations

3.4.10 Create graphics documents using drawing and painting software programs.

3.4.11 Add transitions to slide shows

3.4.12 Add special effects to a presentation

3.4.13 Run slide shows manually and automatically

3.4.14 Save slide show presentations

Unit 3.5: Databases: Microsoft Access

3.5.1 Identifying the objects on the Microsoft Access design window

3.5.2 Creating tables including field definition, setting input masks, default values, validation rules, and selection the primary key

3.5.3 Establishing one-to-one and one-to-many relationships between tables

3.5.4 Designing forms using various types of controls

3.5.5 Creating select, update, and append queries

3.5.6 Performing calculations using queries

3.5.7 Creating reports

3.5.8 Printing information.

Unit 3.6: Desktop Publishing: Microsoft Publisher

3.6.1 Preparing publications using desktop publishing software
3.6.2 Formatting new desktop publishing files
3.6.3 Entering information directly into document
3.6.4 Placing preformatted text into document
3.6.5 Placing graphics in document
3.6.6 Employing draw boxes
3.6.7 Creating graphics files using clip art
3.6.8 Importing scanned files
3.6.9 Enhancing publications using different fonts, styles, attributes & justification.
3.6.10 Enhancing publications using paint/draw functions
3.6.11 Creating two-sided documents
3.6.12 Performing editing functions
3.6.13 Setting up master pages
3.6.14 Outputting desktop publishing files

Unit 3.7: Sending and Receiving E-mail: Outlook Express

3.7.1 Identifying elements of Outlook Express.

3.7.2 Composing e-mail.

3.7.3 Sending e-mail.

3.7.4 Receiving e-mail.

3.7.5 Replying to e-mail.

3.7.6 Forwarding e-mail.

3.7.7 Observing ‘netiquette’ when using e-mail.

3.7.8 Describing various Outlook Express folders.

Unit 3.8: Using the Internet: Microsoft Explorer

3.8.1 Explain internet and web

3.8.2 Explain the benefits of using internet

3.8.3 Browse the web

3.8.4 Open web pages

3.8.5 Save web pages

3.8.6 Use search engines

3.8.7 Download documents

3.8.8 Apply rules for using the internet

3.8.9 Explain Video and Teleconferencing

RECOMMENDED READING

Gilbert, R..(2002). *Getting Things Done On Your PC.* London: Dorling Kindersley (DK)

MODULE 4 COMPUTERISED SWITCH BOARD OPERATIONS

Nominal Duration: 138 hours

Aim: The module aims at providing trainees with skills and competencies required to effectively and efficiently carry out switchboard operations.

Objectives: At the end of the Module trainees will be able to:

1. Describe the Safety and Security Act
2. Operate a computerised switchboard effectively and efficiently.
3. Operate a telephone system effectively and efficiently.
4. Make various types of phone calls.

Unit 4.1 Safety and Security

4.1.1 Shops, Offices and Railway Act (1963)
4.1.2 Health and Safety Act (1974)
4.1.3 Information

Unit 4.2 Types of Switchboards

4.2.1 Computational and Non- Computational Switchboards
4.2.2 Private Branch Exchange (PBX) Characteristics
4.2.3 Private Automatic Branch Exchange (PABX)
4.2.4 Private Manual Branch Exchange (PMBX)
4.2.5 Advantages and Disadvantages of using Computerised and Non-Computerised Switchboards

Unit 4.3 Office Telephone Systems

4.3.1 Different Office Telephone Systems
4.3.2 Operation of Cellular Phones

Unit 4.4 Types of Calls and Making Phone Calls

4.4.1 Different types of telephone tones
4.4.2 Characteristics of a Local, Trunk and International Call
4.4.3 Making Types of Telephone Calls
4.4.4 Answering Telephone Calls
4.4.5 Taking Telephone Messages

Unit 4.5 The Receptionist

4.2.1 Physical Location of a Receptionist Desk
4.2.2 Main Duties of a Receptionist
4.2.3 Qualities of a Good Receptionist
4.2.4 Types of Clients

Unit 4.6 Telephone Operating

4.6.1 Different Types of Telephones
4.6.2 Qualities of a Good Telephone Operator
4.6.3 Identifying faults and available support
4.6.4 Troubleshooting and Help Systems

MODULE 5 COMPUTERISED BRAILLE

Nominal Duration: 252 hours

Aim: This module aims at providing trainees with knowledge and skills in Braille production.

Objectives: At the end of the Module, trainees will be able to

1. Scan Braille documents.
2. Process Braille documents.
3. Format Braille documents.
4. Import and translate Braille documents.
5. Emboss Braille documents.
6. Bind and label Braille documents.

Unit 5.1 History of Braille Production

5.1.1 History of Braille production
5.1.2 Significant advances in Braille production
5.1.3 Scanners and Braille printers
5.1.4 Major milestones in the development of Braille production

Unit 5.2 Hardware Components Associated with Computerised Braille

5.2.1 Classification of scanners e.g. OCR
5.2.2 Classification of embossers e.g. Braille printers
5.2.3 Classification of assistive devices
5.2.4 Calibration of Paper Sensor

Unit 5.3 Software Components Associated with Computerised Braille

5.3.1 Installing Win Braille
5.3.2 Installing Duxbury
5.3.3 Features of Win Braille and Duxbury
5.3.4 The role of Win Braille and Duxbury in Computerised Braille
5.3.5 Installing software for scanners
5.3.6 Features of Omni Page and Text Bridge Millennium
5.3.7 The role of Omni Page and Text Bridge Millennium in Computerised Braille
5.3.8 Creating User Profiles

Unit 5.4 Starting Your First Document

5.4.1 Opening Duxbury Braille Translator
5.4.2 Opening a new document in Duxbury
5.4.3 Entering new text in the document
5.4.4 Translating a document into Braille
5.4.5 Embossing the document
5.4.6 Saving and closing the document

5.4.7 Exiting Duxbury Braille Translator Window

Unit 5.5 Working with Existing Print Document in Win Braille and Duxbury

5.5.1 Opening an existing print document
5.5.2 Editing an existing print document
5.5.3 Saving an existing print document
5.5.4 Translating an existing print document into Braille
5.5.5 Editing an existing Braille document
5.5.6 Embossing
5.5.7 Saving and closing a document

Unit 5.6 Formatting Documents

5.6.1 Differences between Print and Braille Documents
5.6.2 Formatting a document in Microsoft Word
5.6.3 Importing a document to Duxbury
5.6.4 Formatting a document in Duxbury Braille Translator
5.6.5 Translating a document into Braille
5.6.6 Embossing
5.6.7 Saving and closing a document

Unit 5.7 Career Pathways, Security Risks and Associated Safeguards

5.7.1 Computer literacy effects on productivity in a work place
5.7.2 Computers empowering employees
5.7.3 Career development opportunities due to computerisation in industry
5.7.4 Career development implications of becoming computer literate
5.7.5 Security risks and associated safeguards that go with computerisation and using computers

MODULE 6 ENTREPRENEURSHIP

Nominal Duration: 120 HOURS

Aim: The aim of the module is to create awareness of the importance of entrepreneurship and its relevance to career growth by building entrepreneurial competences, attitudinal change that will lead to self motivation and generation of business ideas.

Objectives: At the end of the module trainees should be able to:

1. Identify enterprise management skills.
2. Describe the requirements for establishing an enterprise.
3. Describe the competencies required to sustain an enterprise.

UNIT 6.1: ENTERPRISE MANAGEMENT SKILLS

6.1.1 Managerial and leadership skills in an enterprise
6.1.2 Marketing
6.1.3 Cost and price per product / service
6.1.4 Management of finances
6.1.5 Business records
6.1.6 Enterprise and technology
6.1.7 Business ethics and values in managing an enterprise

UNIT 6.2: ESTABLISHING AN ENTERPRISE

6.2.1 Business Plan/Project
6.2.2 Procedures for formalising an enterprise.

UNIT 6.3: SUSTAINING AN ENTERPRISE

6.3.1 Appraise one's enterprise
6.3.2 Productivity and management
6.3.3 Manage survival and growth
6.3.4 Project

RECOMMENDED READING

1. Calvin, D. R. (2002). *Entrepreneurship Management*.. McGraw Hill

2. Kirby, D (2002). *Entrepreneurship.* Maidenhead: McGraw Hill.

3. Basingstoke, P.B. (2001). *Entrepreneurship & Small Business*. Palgrave.

4. Harper, D.A. (1996). *Entrepreneurship and the Market process: An enquiry into the growth knowledge.*